Yujia Zeng

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Education

University of Science and Technology of China, School of the Gifted Young, Anhui 2023-Present
Undergraduate Student in Artificial Intelligence, Expected to graduate in June 2027
GPA: 3.92 / 4.3 Rank: 1 / 23 (Top 5%)

Publications

1. **Yujia Zeng***, Chensheng Peng*, Yuxin Chen*, Alex Shao, Nathan Jew, and Masayoshi Tomizuka.
RoboSTAR: Next-Scale Autoregressive Sign Language Translation for Humanoid Robots. Received the **Best Paper Award** at the *IROS 2026 Workshop on Nonverbal Cues for Human-Robot Cooperative Intelligence*; *under review at ICLR 2027*. Equal contribution.

2. Bowen Xue, Brandon Y. Feng, Chenguo Lin, Yuchen Lin, **Yujia Zeng**, Lvmin Zhang, Maneesh Agrawala, Honglei Yan, and Panwang Pan.
Ring Forcing: Towards Precise Long-Term Memory for Autoregressive Video Diffusion. *European Conference on Computer Vision (ECCV)*, 2026.

3. **Yujia Zeng**.
ReconNet: Generative Recommendation with Control-Guided Diffusion Models. *International Conference on Intelligent Computing (ICIC)*, 2026. **Oral Presentation**.

4. Ying Yang, Zhengyao Lv, **Yujia Zeng**, Tianlin Pan, Haofan Wang, Yueming Lyu, Binxin Yang, Hubery Yin, Chen Li, Jing Lyu, Ziwei Liu, and Chenyang Si.
StableWorld: Towards Stable and Consistent Long Interactive Video Generation.
arXiv preprint, 2026. **100+ GitHub stars**.

5. Chensheng Peng, Chenpeng Su, Chengwei Zhang, Jiuming Liu, **Yujia Zeng**, Yixiao Wang, Chenfeng Xu, Yichen Xie, Thomas Monninger, Zhipeng Mo, Sihao Ding, Wenzhao Zheng, Kurt Keutzer, Masayoshi Tomizuka, and Wei Zhan.
GenRec4D: Hybrid 4D Gaussian Reconstruction with Feedforward and Generative Priors. *Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*.

6. **Yujia Zeng** et al.
FAR-HOI: From Fine-Grained Human–Object Interaction Generation to Agentic Dexterous Humanoid Manipulation.
Ongoing Research.

Research Experience

UC Berkeley MSC Lab Supervisor: Prof. Masayoshi Tomizuka Mar. 2026 – Mar. 2027
Generative Models, Robotics, Dexterous Hand **J-1 Research Intern**

- Proposed **RoboSTAR**, a next-scale autoregressive framework for speech/text-conditioned sign language motion translation for humanoid robots, achieving state-of-the-art performance.
- Contributed to **GenRec4D**, a hybrid 4D Gaussian reconstruction framework that combines feedforward reconstruction with generative priors for high-quality dynamic scene reconstruction.

Harvard University Embodied Minds Lab Supervisor: Prof. Yilun Du Sep. 2026 – Present
Embodied AI, Dexterous Manipulation **Research Intern**

- Conducting research on fine-grained human–object interaction generation and agentic dexterous humanoid manipulation in collaboration with the MSC Lab at UC Berkeley.

UC Berkeley Collaboration with Prof. Koushil Sreenath Oct. 2026 – Present
Dexterous Humanoid Manipulation, Robot Control **Research Collaboration**

- Collaborating on **FAR-HOI**, integrating fine-grained human–object interaction priors with agentic planning for dexterous humanoid manipulation.

Nanyang Technological University Prof. Ziwei Liu & Chenyang Si Aug. 2025 – Feb. 2026
World Models, Video Generation **Research Project**

- Presented **StableWorld**, investigated long-horizon interactive video generation, focusing on error accumulation and temporal drift in autoregressive world models.
- Developed a model-agnostic dynamic frame eviction mechanism that filters degraded historical frames while retaining geometrically consistent context for more stable long-term generation.

USTC Lab for Data Science Supervisor: Prof. Xiangnan He Aug. 2024-Present
Diffusion, Recommendation Systems **Individual Research Project**

- Designed **ReconNet**, reformulating sequential recommendation as a control-guided diffusion generation problem, using multi-domain user preferences as controllable signals.
- Designed residual-control and cross-norm mechanisms to guide personalized item-embedding generation.

Industry Experience

Baidu Mar. 2025 – Jul. 2025
AIGC, Reinforcement Learning **Algorithm Intern**

- Applied **Direct Preference Optimization (DPO)** using Python and PyTorch to align image-generation models with human preferences for real-world applications.
- Developed and evaluated image-generation workflows, supporting iterative model improvement and product integration.

HiDream Sep. 2025 – Jan. 2026
Generative AI, Image Editing **Algorithm Intern**

- Explored instruction-guided image editing and controllable generation for fine-grained visual manipulation and refinement.
- Built generative-AI workflows using **ComfyUI** for visual content creation, and prototyped website design.

Honors & Awards

- National Scholarship (*top 1% universally acknowledged highest national undergraduate honor*)
- Yang Ya Alumni Fund Scholarship (*top 5 female students in School of the Gifted Young*)
- AI Elite Class Scholarship (*top 3 students in the Artificial Intelligence major*)
- Guangfa Securities Scholarship (*one of 10 recipients recognized for research and competition achievements*)
- ICPC HongKong Regional Contest Silver Medalist (*International Collegiate Programming Contest*)

Skills

- Programming & ML: Python, C/C++, MATLAB, PyTorch, OpenCV, Hugging Face
- Robotics & Simulation: ROS/ROS2, MuJoCo, Isaac Lab, Open3D
- Tools & Systems: Linux, Git, Docker, SLURM, Weights & Biases

Activities

- TOEFL iBT: 5.5/6.0 (Equivalent to 110/120)
- President of the Student Union, School of the Gifted Young
- the Outstanding Physical Fitness Award (top 1% university-wide)